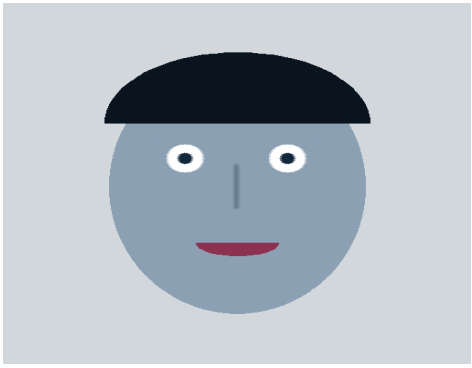
MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

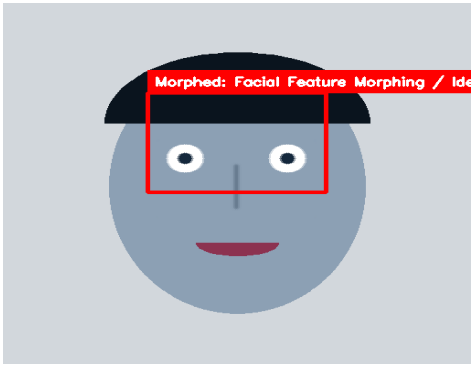
IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	sample_face_fake.png	**Target Domain:**	Face
Audit Timestamp:	2026-09-23 11:59:42 UTC	**SHA-256 Hash:**	8fc6470667c947dc6aedeb0f...
Forensic Verdict:	**MANIPULATED / DEEPFAKE DETECTED**	**Confidence:**	**93.6%**
Tamper Type:	Facial Feature Morphing / Identity Swap	**Bounding Box:**	[158, 127, 353, 268]

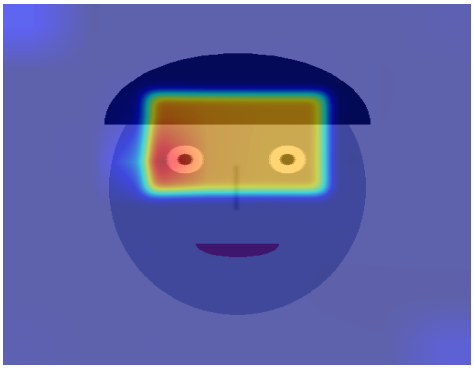
Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.9445 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 3.1733
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.908
MEFCN (Multi-Expert Consensus)	{'Face': 93.6, 'Medical Radiograph': 0.9, 'Satellite Terrain': 2.4, 'Regular Out-of-Domain': 3.2, 'Natural Scene': 2.4}
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 1/1 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 2.09 | Sensor Noise Var: 384.3

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.